

Diet Analyzer

Javier Gutierrez Back & George Chaidemenos

2024-04-23

Table of contents

Introduction	1
Wrangling	3
Exploratory Analysis	7
Findings and Results	7
Conclusion	7
References	8

Introduction

- Title: Diet Analyzer
- Purpose:

Describe the general topic/phenomenon you want to explore, as well some carefully considered questions that you hope to address. You should make an argument motivating your work. Why should someone be interested in what you are doing? What do you hope people will learn from your project?

We want to make it easy for people to visualise what they ate in a particular day in terms of their nutrients intake, and somehow quantify how ‘balanced’ their diet was. One question we would like to answer is what proportion of a person’s meal was in each food group. Additionally, we would like people to be able to input to the program what they ate in the meal and have it say the amounts of each nutrient that they had. People would care about our project since it would be easier for them to find patterns in their eating habits and how to make it healthier perhaps or help them plan meals in advance.

- Data:

As best you can, describe where you will find your data, and what kind of data it is. Will you be working with spatial data in shapefiles? Will you be accessing an API to a live data source? Be as specific as you can, listing URLs and file formats if possible.

We found our data set in the data is plural website. It is called Canadian Nutrient and it is a relational data base that provides information about food descriptions, food groups, nutrients, and a method for calculating the amounts of these nutrients for different quantities of the food. The data set contains 12 tables in a csv format which were already documented and loaded as an R package.

We will not be working with spatial data in shapefiles, or using an API to a live data source. (<https://www.canada.ca/en/health-canada/services/food-nutrition/healthy-eating/nutrient-data/canadian-nutrient-file-2015-download-files.html>)

- Variables:

List, and briefly describe, each variable that you plan to incorporate. If you can, be specific about units, scale, etc.

- food: type of food
- nutrient: name of the nutrient
- nutrient unit: the unit of measure (g, mg)
- nutrient quantity: nutrient quantity in nutrient unit per 100 grams of the food
- measure name: type of quantity used to measure the food
- conversion factor value: the factor by which we multiply the nutrient quantity per the measure used
- food group: the general category the group belongs to
- End Product:

Describe what you hope to deliver as a final product. Will it be a Shiny application that will be posted on the Internet (I anticipate that most groups will incorporate some form of dynamic visualization)? Will it be a GoogleMaps mash-up? Will it be an R package that provides an API to a live data source? Will it be an exploration of an interesting network dataset?

We would hope to create a user interactive app through shiny where the user will input what foods they had eaten in a day and the app will provide the user with an analysis of their diet for that particular day. For example, create a pie chart or other visualisation that shows what proportions of different food groups they consumed, and a detailed analysis of how much fat, protein or other specific nutrients they consumed. We would like the user to be able to specify

the specific foods and quantities as closely to the ones on the data set to provide accurate data with the conversion factors. For both of these questions, we would like to compare the user's numbers with certain baseline diets, like an athlete, a bodybuilder, a health nut, keto diet, vegan, maybe gluten free, fast food enthusiast. That is, showing the food group proportions of for the corresponding diets, do some sort of calculation to see which one the user is closest too, and maybe provide sample diets or possible improvements to make the diet more balanced or to satisfy your objective diet requirements.

For the Shiny app, we will probably like to have an 'input your meal' section, where the user can type in their foods and match them (with regular expressions in the back end) to the foods in the database. From there, you will see two sections, one with the food group proportion graph, and one with your nutrient intake in descending order. We could still figure out whether we want the baseline diet comparison in the same sections, or if we want another section below those with possible changes and the comparisons.

- Do you have any questions for me?

Wrangling

```
# if things go quickly, we might wrangle data from `data-raw` into `data` here.
# these would be referred to as `../data-raw` and `../data`, respectively.
# if not, the wrangling should happen in a `wrangle.qmd` file here (with associated pdf).

# This meal would come from the search bar and the user's input. We are using a random meal
Meal <- FoodNames |>
  filter(food_description == 'Cheese souffle' | food_description == 'Beef pot roast, with
  inner_join(ConversionFactor, by = 'food_id') |>
  inner_join(MeasureNames, by = 'measure_id') |>
  filter(measure_description == '250ml' | measure_description == '140g' | measure_descripti

Meal |> select(food_description, measure_description)
```

```
# A tibble: 3 x 2
  food_description                measure_description
  <chr>                          <chr>
1 Cheese souffle                 250ml
2 Beef pot roast, with browned potatoes, peas and corn 140g
3 Chicken, broiler, light meat, fried 100ml chopped or diced
```

```
# Wrangling to get the value of each nutrient for our meal and select the nutrients that w

analyse_meal <- function(Meal) {
  Nutrients <- Meal |>
  inner_join(NutrientAmounts, by = 'food_id') |>
  inner_join(NutrientNames, by = 'nutrient_id') |>
  filter(nutrient_name == 'PROTEIN' | nutrient_name == 'FAT (TOTAL LIPIDS)' | nutrient_name == 'CARBOHYDRATE, TOTAL (BY DIFFERENCE)' | nutrient_name == 'SUGARS, TOTAL') |>
  mutate(new_value = nutrient_value * conversion_factor_value) |>
  group_by(nutrient_name) |>
  summarise("value" = sum(new_value)) |>
  select(nutrient_name, value) |>
  arrange(desc(value))

  return (Nutrients)
}

Nutrients <- analyse_meal(Meal)

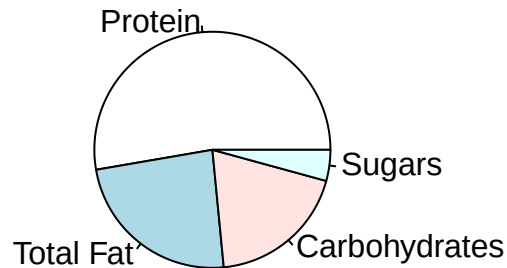
Nutrients
```

```
# A tibble: 4 x 2
  nutrient_name      value
  <chr>            <dbl>
1 PROTEIN          58.1
2 FAT (TOTAL LIPIDS) 26.3
3 CARBOHYDRATE, TOTAL (BY DIFFERENCE) 21.2
4 SUGARS, TOTAL     4.69
```

```
# Pie chart showing the proportions of those nutrient in our meal.

labels = c("Protein", "Total Fat", "Carbohydrates", "Sugars")
# Vitamins and minerals to be added later
pie(Nutrients$value, labels, main = "Diet Nutrient Proportions")
```

Diet Nutrient Proportions



```
# Count items in each food group, will find a better way to quantify this
# Would look better with an actual meal rather than three items
```

```
OurFoodGroups <- Meal |>
  inner_join(FoodGroup, by = 'food_group_id') |>
  group_by(food_group_name) |>
  summarise("Number of items" = n())
```

```
OurFoodGroups
```

```
# A tibble: 2 x 2
  food_group_name `Number of items`
  <chr>           <int>
1 Mixed Dishes      2
2 Poultry Products  1
```

```
# To help us quantify nutrients based on how much of the food the user ate.
```

```
NutrientNames |>
  inner_join(NutrientAmounts, by = 'nutrient_id') |>
  rename("vitamin_b12_in_g" = nutrient_value) |>
  filter(nutrient_name == 'VITAMIN B-12') |>
  inner_join(FoodNames, by = 'food_id') |>
```

```

filter(food_description == 'Egg, chicken, whole, cooked, scrambled or omelet') |>
inner_join(ConversionFactor, by = 'food_id') |>
inner_join(MeasureNames, by = 'measure_id') |>
mutate(total_vitamin_b12_in_g = vitamin_b12_in_g * conversion_factor_value) |>
select(measure_description, conversion_factor_value, total_vitamin_b12_in_g)

```

```
# A tibble: 6 x 3
```

	measure_description	conversion_factor_value	total_vitamin_b12_in_g
	<chr>	<dbl>	<dbl>
1	1 large egg	0.61	0.853
2	100ml	0.930	1.30
3	250ml	2.32	3.25
4	1 small egg	0.48	0.671
5	110g	1.1	1.54
6	2 large eggs	1.22	1.71

```
#baseline diets
```

```
athlete <- tibble(
```

```
  nutrient_name = c("PROTEIN", "FAT (TOTAL LIPIDS)", "CARBOHYDRATE, TOTAL (BY DIFFERENCE)"
  proportions = c(20.0, 30.0, 50.0)

```

```
)
```

```
normal <- tibble(
```

```
  nutrient_name = c("PROTEIN", "FAT (TOTAL LIPIDS)", "CARBOHYDRATE, TOTAL (BY DIFFERENCE)"
  proportions = c(15.0, 27.5, 57.5)

```

```
)
```

```
vegan <- tibble(
```

```
  nutrient_name = c("PROTEIN", "FAT (TOTAL LIPIDS)", "CARBOHYDRATE, TOTAL (BY DIFFERENCE)"
  proportions = c(17.5, 27.5, 55.0)

```

```
)
```

```
fastfoodlover<- tibble(
```

```
  nutrient_name = c("PROTEIN", "FAT (TOTAL LIPIDS)", "CARBOHYDRATE, TOTAL (BY DIFFERENCE)"
  proportions = c(15.0, 37.5, 47.5)

```

```
)
```

```
carnivore <- tibble(
```

```
  nutrient_name = c("PROTEIN", "FAT (TOTAL LIPIDS)", "CARBOHYDRATE, TOTAL (BY DIFFERENCE)"
  proportions = c(30, 52.5, 17.5)

```

```
)
```

```
desertlover <- tibble(  
  nutrient_name = c("PROTEIN", "FAT (TOTAL LIPIDS)", "CARBOHYDRATE, TOTAL (BY DIFFERENCE)"  
  proportions = c(7.5, 32.5, 60.0)  
)
```

Exploratory Analysis

```
# we will ingest the analytic data from `data` here (as `../data`).
```

Findings and Results

Conclusion

We probably don't want to cite Broman and Woo (2018), but if we did this is how we would do it. Or we might talk about the importance of organizing data in spreadsheets (Broman and Woo 2018) and provide a different format for the citation.

References

Broman, Karl W., and Kara H. Woo. 2018. “Data Organization in Spreadsheets.” *The American Statistician* 72 (1): 2–10. <https://doi.org/10.1080/00031305.2017.1375989>.